

Wright brothers lied

Newly discovered evidence could suggest that the honour of first flight could go to a German immigrant: <http://dgit.in/ZhNmbw>



Miniscule memory

Scientists are working on a way to use one atom thick graphene as flash storage. <http://dgit.in/14co3zv>

Headphones Test

HOW WE TESTED

Audiophile-grade: For testing these audio pieces we got the ASUS Xonar Essence One headphone DAC and amp. Since we didn't know which headphones will find their way into the test, we were expecting high impedance sets that are really difficult to drive. Secondly, we wanted the equipment to be as neutral as possible and both the Essence One and STX (our backup) are supposed to be as neutral as you can get in that price range. We did not use bit perfect or any software or hardware level upsampling. The other part of the puzzle was getting the right test tracks of the highest possible quality. Of course we used only fully uncompressed WAV files or tracks

encoded in lossless FLAC. The sampling rate was mostly 44.1 kHz but we did have a few 192 kHz tones as well. But limited to non-optical primary source we kept the DAC by default to 44.1 kHz. As for track selection we used a series of tracks (some of which we've been using for years now) to test specific attributes such as stereo separation, vocal tonality, detail (when instrument density is high). As most of you know, we have a penchant for guitar driven music, but for this test we've diversified our genres. Apart from these performance tests the build, comfort and features were rated. Longer, woven cables for instance got a better score. We didn't score

attributes such as isolation, or the ability to fold the headphones as they don't take away from the listening experience.

Wireless portable: To be honest, the wireless portable category formed as a matter of chance. We sent out requests for audiophile grade headphones and since many brands sent us wireless sets we decided to club them in. Unfortunately since most of these headphones didn't have aux inputs our amp was rendered useless. We had to depend on low fidelity Bluetooth, which was in fact a realistic use case test. The rest was same except points were given for the ability to fold and wireless range.

colouring, if at all needed should be done externally. That's what DSP effects are for!

So far if you've noticed we haven't used the word "Audiophile". While the word carries a lot of meaning, it's somehow been overused and almost abused of late. Every other manufacturer is coming out with Audiophile speakers or headphones, which in all honesty can hardly even qualify as decent audio, let alone Audiophile-grade. At the same time, there are self-proclaimed Audiophiles (especially on forums) who like to try as many headphones as they can get their hands on but derive little joy from them except laying a claim to a misplaced sense of elitism. Nevertheless, an offshoot of all this newfound interest in audio is that the overall level of the industry as well has the listener has risen. There's literally a legion of true music enthusiasts who want to get the best audio equipment for their given budget and are willing to understand the various components of what good audio is all about. This is where we come in. But our job isn't easy.

Describing sound with words is perhaps one of the biggest

challenges a writer or reviewer will ever face. Most will resort to using ambiguous adjectives ranging from woolly bass to sinewy tones. We for certain have yet to encounter woolly bass or sinewy tones in all our listening years. The only alternative to avoid a subjective opinion or flowery adjectives is to empirically analyse the audio signal reproduced by the headphones. How do you do that? By letting instruments do the listening – play an audio sample, record it via a microphone, and analyse the resulting frequency curve to check whether it is true to source. The last time we did this for one of our speaker tests, we didn't get much variation in our results. The reason was the instruments we were using weren't sophisticated enough. So until we get our hands on the perfect audio testing set up – custom moulded maniquin head for binaural placement, high end condenser microphone etc – we rely on our organic god given instruments: our good old wizened ears. Why should you trust anyone's ears, you may ask? Well one of the qualifying factors is the sheer number of headphones (or any audio-gear for that matter) that has passed through them at our

test center. But beyond that we rely on a lot of testing standards to ensure as little ambiguity and reviewer's bias as possible. Have a look at our "How we tested" boxes before proceeding to the actual test results.

AUDIOPHILE HEADPHONES Above ₹12,000

When we sent out our sourcing mails to various manufacturers and distributors we didn't specify a lower or higher limit. Expectedly, we got some really awesome gear. In this first price based category we got some very interesting entrants. From the top of the line Sennheiser HD 700 to some interesting cans from the stables of Logic3 with their Ferrari branding and noise cancelling features. There were also some unexpected and almost serendipitous packages that found their way to our doorsteps. The first was the Martin Logan Mikros-90, which is perhaps the first headphone from a brand famous for electrostatic speakers. The second came as a result of a rather well timed press event arranged by Blaupunkt. After the entire testing was done we got hold of two rather interesting head-

phones from the German audio manufacturer. After a night of overnight testing we managed to squeeze them both into this test. If you've got the magazine late this time you know who to thank. Of course we kid, but on that note here's a quick and dry run down of this category:

Logic3 Ferrari T350

This was initially the only active noise cancelling piece in the entire test until it was joined by the Blaupunkt CN-112 Comfort. The ANC works well enough to cut out ambient hum but strangely adds its own buzz; though barely noticeable, can get very annoying if you're not used to it. As with some of the ANC headphones nowadays they don't work without the ANC on. Often you forget to switch off the ANC and in this case the headphone. We ran through two sets of AAA Duracell batteries during the course of this test. Apart from actual play time, a lot of it was just simply because the headphones were left "on". With a nominal impedance rated at 32 ohms and being highly sensitive, their easy to drive and loud as hell. Unlike the HD700, which manages to create a per-